Tutorial Proposal

Title: Statistical Learning Methods for Emerging Database Applications

Target Audience

Broad audience including students and researchers interested in an overview of Statistical Learning algorithms for analyzing high-dimensional data such as imagery and gene expression. No particular background expected.

Motivation and Goals

Statistical Learning is a well-established scientific discipline. The theories underlying classical Statistical Learning are based on the assumption of $D < N$ and $N \rightarrow \infty$, where D denotes the number data dimensions and N the number of training instances. Many emerging data-analysis applications, however, face the $D > N$ high dimensionality challenge. For instance, an image/video search engine needs to learn users' query concepts with a very small number of training instances (provided by users via relevance feedback) in very high-dimensional feature spaces. In gene-expression profiling, each human cell contains approximately three billion base gene-pairs, which encode between $50,000$ to $100,000$ genes, but the available training data for analyzing a particular genetic disease is typically fewer than a hundred.

This tutorial will present statistical methods for making inferences in very high dimensional spaces. This four-hour tutorial is organized into three parts.

Outline

1. *Classical methods*. (1 hour)
- Statistical decision theories.
- The least squares, the nearest neighbors, and anything in between.
- Ensembles: bagging, boosting, random forests.

2. *Kernel methods*. (2 hour)
- Gaussian model.
- Support Vector Machines.
- Bayes Point Machines.
- Kernel functions and parameter selection.

3. *High dimensional inferencing*. (1 hour)
- Blessings of high dimensionality.
- Curses of high dimensionality.
- Dimension reduction techniques.

Tutor

Prof. Edward Chang received his Ph.D. in Electrical Engineering at Stanford University in 1999. He is an Assistant Professor of Electrical and Computer Engineering at the University of California, Santa Barbara. His research interests include statistical learning and multimedia databases. He is a recipient of the IBM Faculty Partnership Award from 2000 to 2002, and the NSF Career Award in 2002. His perception-based image retrieval (PBIR) work, which applies statistical learning to understand users' subjective similarity metrics, was recognized as a major breakthrough in image retrieval by the CBIR panel at the 2002 IEEE International Conference on Multimedia. Prof. Chang is a co-founder and the CTO of VIMA Technologies.